3.1.2 Loads acting on chain and belt shafts

In the case of power transmission by a chain and a belt, the load acting tangentially on the sprocket or the pulley can be determined by **formula (3.7)**.

$$K_t=\frac{19.1\times10^6\cdot HP}{D_p\cdot n}\left(=\frac{1.95\times10^6\cdot HP}{D_p\cdot n}\right)\cdots\cdots\cdots\cdots\cdots\cdots\textbf{(3.7)}$$

where,

K_t : Tangential load acting on sprocket or pulley N (kgf)
HP : Transmission power kW
D_p : Pitch circle diameter of sprocket or pulley mm

In the case of belt driving, initial tension is applied to the belt so the pulley and the belt are both always pressed down with a proper load.

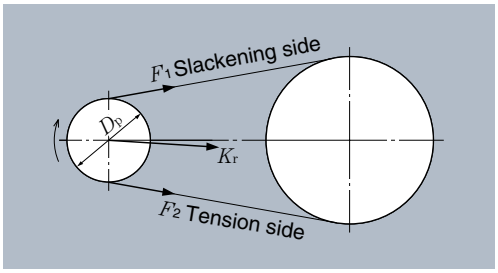


Fig. 3.6 Loads acting on chain/ belt

Considering this initial tension, radial load acting on the pulley is expressed in **formula (3.8)**.

In the case of chain driving, the radial load acting thereon can be expressed using the same formula, if vibration and shock are taken into consideration.

$$K_r=f_b\cdot K_t\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\textbf{(3.8)}$$

where,

K_r : Radial load acting on sprocket or pulley N (kgf)
f_b : Chain/belt factor **(Table 3.3)**

Table 3.3 Chain/belt factor f_b

Type of chain / belt	f_b
Chain (single row type)	1.2～1.5
Vee-belt	1.5～2.0
Timing belt	1.1～1.3
Flat belt (with tension pulley)	2.5～3.0
Flat belt	3.0～4.0

3.1.3 Load factor

In an actual machine, the shaft load is mostly greater than the theoretically calculated load due to vibration, shock, etc. The load acting on the shafts of a machine can be determined by **formula (3.9)**.

$$K=f_w\cdot K_C\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\textbf{(3.9)}$$

where

K : Actual load acting on shaft N(kgf)
K_c : Theoretically calculated value N (kgf)
f_w : Load factor **(Table 3.4)**

Table 3.4 Load factor f_w

Extent of shock	f_w	Application
Nearly no shock	1.0～1.2	Electrical machines, machine tools, measuring instruments
Light shock	1.2～1.5	Railway vehicles, automobiles, rolling mills, metal working machines, paper making machines, rubber mixing machines, printing machines, aircraft, textile machines, electrical units, office equipment
Heavy shock	1.5～3.0	Crushers, agricultural machines, construction machines, cranes

3.2 Load distribution to bearings

Any loads acting on shafts are distributed to the bearings considering static tension to be supported with the bearings.

For example, the loads acting on the bearings in the gear shaft illustrated in **Fig. 3.7** can be expressed in **formulas (3.10)** and **(3.11)**.

$$F_{rA}=K_{r\,I}\frac{b}{l}-K_{r\,II}\frac{c}{l}-K_a\frac{D_p}{2l}\cdots\cdots\cdots\cdots\cdots\cdots\textbf{(3.10)}$$

$$F_{rB}=K_{r\,I}\frac{a}{l}+K_{r\,II}\frac{a+b+c}{l}+K_a\frac{D_p}{2l}\cdots\cdots\cdots\cdots\textbf{(3.11)}$$

where,

F_{rA} : Radial load acting on bearing-A N (kgf)
F_{rB} : Radial load acting on bearing-B N (kgf)
$K_{r\,I}$: Radial load acting on gear-I N (kgf)
K_a : Axial load acting on gear-I N (kgf)
$K_{r\,II}$: Axial load acting on gear-II N (kgf)
D_p : Pitch circle diameter of gear-I mm
l : Bearing to bearing distance mm

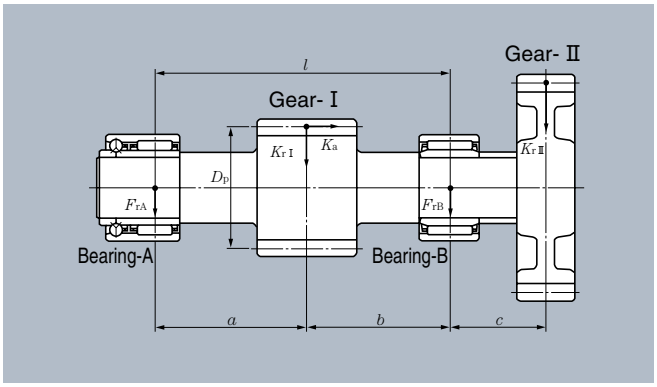


Fig. 3.7 Gear shaft

3.3 Mean load

The loads acting on bearings used in ordinary machines fluctuate mostly depending on the predetermined operation cycle or work plan. In such a case, the mean load F_m converted so the same lifetime is assigned to the bearings is used as the bearing load.

(1) Stepped fluctuating load

Where bearing loads F_1 , F_2 ..... F_n act on and the rotational speed and running time are n_1, n_2.....n_n and t_1, t_2..... t_n respectively, the mean load F_m of these loads can be expressed in **formula (3.12)**.

$$F_m = \left\{ \frac{\Sigma\,(F_i^{\,p}\, n_i\, t_i)}{\Sigma\,(n_i\, t_i)} \right\}^{1/p} \cdots\cdots (3.12)$$

where:

$p = 10/3$ for roller bearing
$p = 3$ for ball bearing

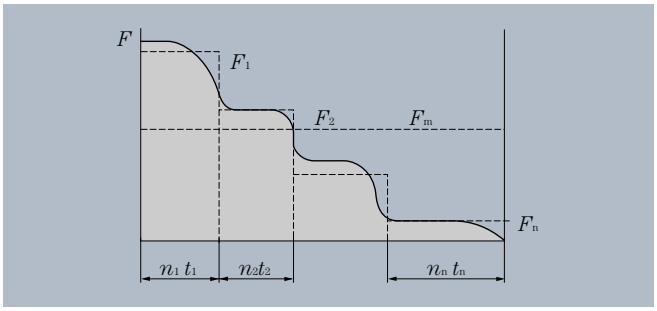


Fig. 3.8 Stepped fluctuating load

(2) Consecutive series load

Where load is load cycle "t_o" and it can be expressed in the function $F(t)$ of time t, the mean load can be expressed in **formula (3.13)**.

$$F_m = \left\{ \frac{1}{t_o} \int_0^{t_o} F\,(t)^p\, dt \right\}^{1/p} \cdots\cdots (3.13)$$

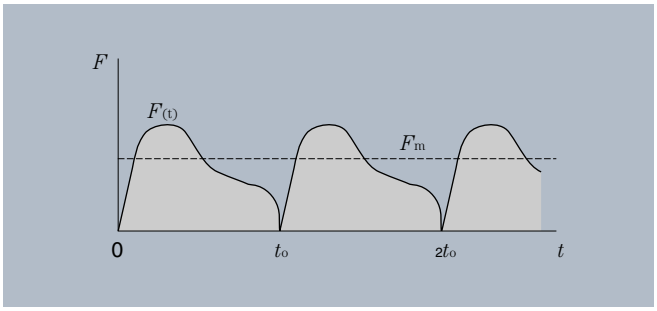


Fig. 3.9 Load fluctuating as a time function

(3) Load fluctuating in nearly linear form

The mean load F_m can be determined approximately by **formula (3.14)**.

$$F_m = \frac{F_{min} + 2F_{max}}{3} \cdots\cdots (3.14)$$

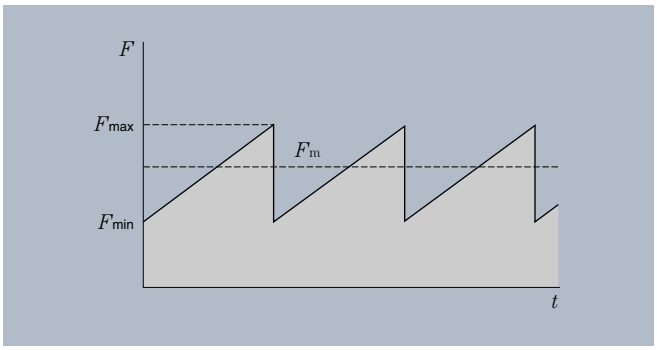


Fig. 3.10 Load fluctuating in linear form

(4) Sinusoidal fluctuating load

The mean load F_m, can be determined approximately by **formulas (3.15)**, **(3.16)**.

case of (a) $F_m = 0.75F_{max}$ ····· **(3.15)**
case of (b) $F_m = 0.65F_{max}$ ····· **(3.16)**

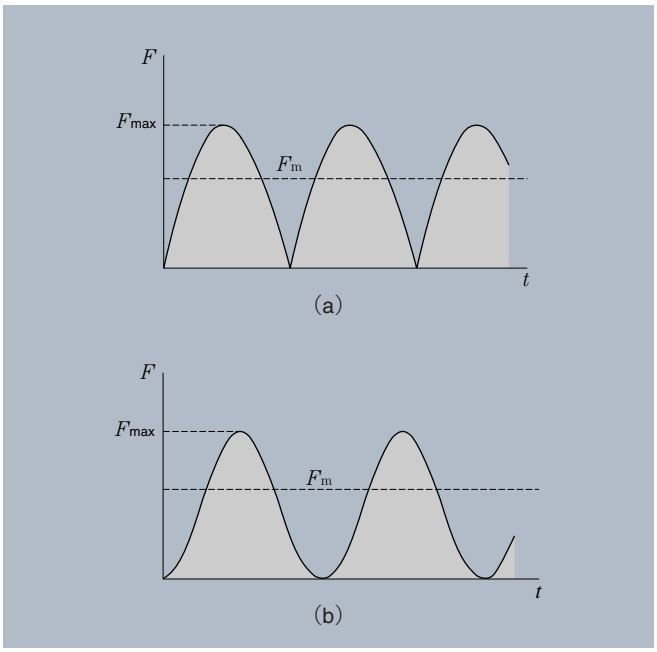


Fig. 3.11 Sinusoidal fluctuating load



4. Bearing Accuracy

The dimensional, profile and running accuracies of rolling bearings are specified in ISO Standard as applicable and JIS B 1514 (Accuracy of Rolling Bearings).

"**Dimensional accuracy**" and "**Profile accuracy**" are the items indispensable in installing the rolling bearings on a shaft and in a bearing housing, and allowable bearing run-out in running is specified as the running accuracy.

Dimensional accuracy:

Dimensional accuracy means the respective allowable values for bore diameter, outer diameter, width or height (limaited to thrust bearing) and chamfering dimension.

Profile accuracy:

This means the allowable value for variation of each of bore diameter, mean bore diameter, outer diameter, mean outer diameter, and width or thickness of bearing ring (limited to thrust bearing).

Running accuracy:

This means the respective allowable values for radial run-out and axial run-out of both inner ring and outer ring, lateral run-out of inner ring, and outer diameter run-out of outer ring.

Regarding the accuracy class of the machined ring needle roller bearings, class-0 is equivalent to bearings of the normal precision class, and precision becomes progressively higher as the class number becomes smaller; i.e. Class 6 is less precise than Class 5, which is less precise than Class 4, and so on.

Bearings of Class-0 are mostly used for general application, while bearings of Class-5 or Class-4 are used, where the required running accuracies and revolutions are high or less friction and less fluctuation are required for bearings.

Various bearing types are available for **NTN** needle roller bearings and the representative types and the accuracy classes applicable to them are as shown in **Table 4.1**.

Dimensional item symbols used in the accuracy standard are given in **Table 4.2**, the radial bearing accuracy specified every accuracy class given in **Table 4.3**, the thrust bearing accuracy specified every accuracy class given in **Table 4.4**, and the allowable values for chamfering dimension given in **Table 4.5**.

Table 4.2 Dimensional item symbols used in applicable standards

Classification	Symbols	Symbol representation	Symbols under JIS B 0021 (Reference)
Dimensional accuracy	Δd_{mp}	Dimensional tolerance for in-plane mean bore diameter	———
	Δd_s	Dimensional tolerance for bore diameter	———
	ΔD_{mp}	Dimensional tolerance for in-plane mean outer diameter	———
	ΔD_s	Dimensional tolerance for outer diameter	———
	ΔB_s	Dimensional tolerance for inner ring width	———
	ΔC_s	Dimensional tolerance for outer ring width	———
Profile accuracy	V_{dp}	Variation of in-plane bore diameter	Roundness ◯ 1)
	V_{dmp}	Variation of in-plane mean bore diameter	Cylindricality ⌭ 2)
	V_{Dp}	Variation of in-plane outer diameter	Roundness ◯ 1)
	V_{Dmp}	Variation of in-plane mean outer diameter	Cylindricality ⌭ 2)
	V_{Bs}	Variation of inner ring width	Parallelism //
	V_{Cs}	Variation of outer ring width	Parallelism //
Running accuracy	K_{ia}	Radial run-out of inner ring	Run-out ↗
	K_{ea}	Radial run-out of outer ring	Run-out ↗
	S_{ia}	Axial run-out of inner ring	———
	S_{ea}	Axial run-out of outer ring	———
	S_d	Lateral run-out (inner ring)	Run-out ↗
	S_D	Outer diameter run-out (outer ring)	Run-out ↗

1) The roundness specified in JIS B 0021 is applicable to the tolerance V_{dp} for variation of radial in-plane bore diameter or nearly half of V_{Dp}.
2) The cylindricality specified in JIS B 0021 is applicable to the tolerance V_{dmp} for in-uniformity of radial in-plane mean diameter or nearly half of V_{Dmp}.

Table 4.1 Bearing types and corresponding accuracy classes

Bearing type		Applicable accuracy class				Applicable table
Needle roller bearing, Clearance-adjustable needle roller bearing		JIS Class-0 —	JIS Class-6 —	JIS Class-5 —	JIS class-4 JIS class-4	**Table 4.3** **Table 4.3**
Complex bearing	Radial bearing Thrust bearing	JIS Class-0 **NTN** Class 0	JIS Class-6 **NTN** Class 6	JIS Class-5 **NTN** Class 5	— **NTN** Class 4	**Table 4.3** **Table 4.4**
Needle roller bearing with double-direction thrust roller bearing	Radial bearing Thrust bearing	— —	— —	JIS Class-5 **NTN** Class 5	JIS Class-4 **NTN** Class 4	**Table 4.3** **Table 4.4**
Thrust roller bearing Roller follower/cam follower		**NTN** Class 0 JIS Class-0	**NTN** Class 6 —	**NTN** Class 5 —	**NTN** Class 4 —	**Table 4.4** **Table 4.3**

Table 4.3 Tolerances for radial bearings
Table 4.3(1) Inner rings

Nominal bore diameter d mm		Dimensional tolerance for mean bore diameter Δ_{dmp}								Variation of mean bore diameter V_{dp}				Allowable variation of bore diameter V_{dmp}			
		Class 0		Class 6		Class 5		Class 4		Class 0	Class 6	Class 5	Class 4	Class 0	Class 6	Class 5	Class 4
over	incl.	high	low	high	low	high	low	high	low	max				max			
2.5❶	10	0	−8	0	−7	0	−5	0	−4	10	9	5	4	6	5	3	2
10	18	0	−8	0	−7	0	−5	0	−4	10	9	5	4	6	5	3	2
18	30	0	−10	0	−8	0	−6	0	−5	13	10	6	5	8	6	3	2.5
30	50	0	−12	0	−10	0	−8	0	−6	15	13	8	6	9	8	4	3
50	80	0	−15	0	−12	0	−9	0	−7	19	15	9	7	11	9	5	3.5
80	120	0	−20	0	−15	0	−10	0	−8	25	19	10	8	15	11	5	4
120	150	0	−25	0	−18	0	−13	0	−10	31	23	13	10	19	14	7	5
150	180	0	−25	0	−18	0	−13	0	−10	31	23	13	10	19	14	7	5
180	250	0	−30	0	−22	0	−15	0	−12	38	28	15	12	23	17	8	6
250	315	0	−35	0	−25	0	−18	—	—	44	31	18	—	26	19	9	—
315	400	0	−40	0	−30	0	−23	—	—	50	38	23	—	30	23	12	—
400	500	0	−45	0	−35	—	—	—	—	56	44	—	—	34	26	—	—

❶ 2.5mm is included in this dimensional category.
❷ This table is applied to the ball bearings.

Table 4.3 (2) Outer rings

Nominal outer diameter D mm		Dimensional tolerance for mean outer diameter Δ_{Dmp}								Allowable variation of outer diameter V_{Dp}				Allowable variation of mean outer diameter V_{Dmp}			
		Class 0		Class 6		Class 5		Class 4		Class 0	Class 6	Class 5	Class 4	Class 0	Class 6	Class 5	Class 4
over	incl.	high	low	high	low	high	low	high	low	max.				max.			
6❸	18	0	−8	0	−7	0	−5	0	−4	10	9	5	4	6	5	3	2
18	30	0	−9	0	−8	0	−6	0	−5	12	10	6	5	7	6	3	2.5
30	50	0	−11	0	−9	0	−7	0	−6	14	11	7	6	8	7	4	3
50	80	0	−13	0	−11	0	−9	0	−7	16	14	9	7	10	8	5	3.5
80	120	0	−15	0	−13	0	−10	0	−8	19	16	10	8	11	10	5	4
120	150	0	−18	0	−15	0	−11	0	−9	23	19	11	9	14	11	6	5
150	180	0	−25	0	−18	0	−13	0	−10	31	23	13	10	19	14	7	5
180	250	0	−30	0	−20	0	−15	0	−11	38	25	15	11	23	15	8	6
250	315	0	−35	0	−25	0	−18	0	−13	44	31	18	13	26	19	9	7
315	400	0	−40	0	−28	0	−20	0	−15	50	35	20	15	30	21	10	8
400	500	0	−45	0	−33	0	−23	—	—	56	41	23	—	34	25	12	—
500	630	0	−50	0	−38	0	−28	—	—	63	48	28	—	38	29	14	—

❶ 6mm is included in this dimensional category.
❷ This table is applied to the ball bearings.

Unit μm

Radial run-out K_{ia}				Face run-out S_d		Axial run-out S_{ia} ❷		Allowable width deviation Δ_{Bs}				Allowable width variation V_{Bs}				Nominal bore diameter d mm	
Class 0	Class 6	Class 5	Class 4	Class 5	Class 4	Class 5	Class 4	Class 0,6		Class 5,4		Class 0	Class 6	Class 5	Class 4		
max				max		max		high	low	high	low	max				over	incl.
10	6	4	2.5	7	3	7	3	0	−120	0	−40	15	15	5	2.5	2.5 ❶	10
10	7	4	2.5	7	3	7	3	0	−120	0	−80	20	20	5	2.5	10	18
13	8	4	3	8	4	8	4	0	−120	0	−120	20	20	5	2.5	18	30
15	10	5	4	8	4	8	4	0	−120	0	−120	20	20	5	3	30	50
20	10	5	4	8	5	8	5	0	−150	0	−150	25	25	6	4	50	80
25	13	6	5	9	5	9	5	0	−200	0	−200	25	25	7	4	80	120
30	18	8	6	10	6	10	7	0	−250	0	−250	30	30	8	5	120	150
30	18	8	6	10	6	10	7	0	−250	0	−250	30	30	8	5	150	180
40	20	10	8	11	7	13	8	0	−300	0	−300	30	30	10	6	180	250
50	25	13	—	13	—	15	—	0	−350	0	−350	35	35	13	—	250	315
60	30	15	—	15	—	20	—	0	−400	0	−400	40	40	15	—	315	400
65	35	—	—	—	—	—	—	0	−450	—	—	50	45	—	—	400	500

Unit μm

Radial run-out K_{ea}				Outside surface inclination S_D		Axial run-out S_{ea} ❹		Allowable width deviation Δ_{Cs}	Allowable width variation V_{Cs}				Nominal outer diameter D mm	
Class 0	Class 6	Class 5	Class 4	Class 5	Class 4	Class 5	Class 4	Class 0,6,5,4	Class 0	Class 6	Class 5	Class 4		
max				max		max			max				over	incl.
15	8	5	3	8	4	8	5	Depending on the tolerance of Δ_{Bs} for d of same bearing.	Depending on the applicable allowable value of V_{Bs} for d of same bearing.		5	2.5	6 ❸	18
15	9	6	4	8	4	8	5				5	2.5	18	30
20	10	7	5	8	4	8	5				5	2.5	30	50
25	13	8	5	8	4	10	5				6	3	50	80
35	18	10	6	9	5	11	6				8	4	80	120
40	20	11	7	10	5	13	7				8	5	120	150
45	23	13	8	10	5	14	8				8	5	150	180
50	25	15	10	11	7	15	10				10	7	180	250
60	30	18	11	13	8	18	10				11	7	250	315
70	35	20	13	13	10	20	13				13	8	315	400
80	40	23	—	15	—	23	—				15	—	400	500
100	50	25	—	18	—	25	—				18	—	500	630

Table 4.4 Tolerances of thrust roller bearings

Table 4.4 (1) Inner rings and center rings

Unit μm

Nominal bore diameter d or d_2 mm		Allowable deviation of mean diameter Δ_{dmp} or Δ_{d2mp}				Allowable variation of bore diameter V_{dp} or V_{d2p}		Allowable variation of raceway thickness ❶ S_i			
		Class 0, 6, 5		Class 4		Class 0, 6, 5	Class 4	Class 0	Class 6	Class 5	Class 4
over	incl.	high	low	high	low	low		max			
—	18	0	−8	0	−7	6	5	10	5	3	2
18	30	0	−10	0	−8	8	6	10	5	3	2
30	50	0	−12	0	−10	9	8	10	6	3	2
50	80	0	−15	0	−12	11	9	10	7	4	3
80	120	0	−20	0	−15	15	11	15	8	4	3
120	180	0	−25	0	−18	19	14	15	9	5	4
180	250	0	−30	0	−22	23	17	20	10	5	4
250	315	0	−35	0	−25	26	19	25	13	7	5
315	400	0	−40	0	−30	30	23	30	15	7	5
400	500	0	−45	0	−35	34	26	30	18	9	6
500	630	0	−50	0	−40	38	30	35	21	11	7

❶ The complex bearings are applicable to the category of single plane bearing d which corresponds to the same nominal outer diameter of same diameter series, without being applicable to d_2 category.

Table 4.4 (2) Outer rings

Unit μm

Nominal outer diameter D mm		Allowable deviation of mean outer diameter Δ_{Dmp}				Allowable variation of outer diameter V_{Dp}		Allowable variation ofraceway thickness S_e
		Class 0,6,5		Class 4		Class 0,6,5	Class 4	Class 0,6,5,4
over	incl.	high	low	high	low	max		max
10	18	0	−11	0	−7	8	5	Depending on the applicable allowable value of S_1 for d or d_2 of same bearing.
18	30	0	−13	0	−8	10	6	
30	50	0	−16	0	−9	12	7	
50	80	0	−19	0	−11	14	8	
80	120	0	−22	0	−13	17	10	
120	180	0	−25	0	−15	19	11	
180	250	0	−30	0	−20	23	15	
250	315	0	−35	0	−25	26	19	
315	400	0	−40	0	−28	30	21	
400	500	0	−45	0	−33	34	25	
500	630	0	−50	0	−38	38	29	
630	800	0	−75	0	−45	55	34	

Table 4.5 Allowable critical value for chamfering dimension

Table 4.5 (1) Radial bearings

Unit mm

rs min❶	Nominal bore diameter d		Radial direction	Axial direction
	over	incl.	rs max	
0.15	—	—	0.3	0.6
0.2	—	—	0.5	0.8
0.3	—	40	0.6	1
	40	—	0.8	1
0.6	—	40	1	2
	40	—	1.3	2
1	—	50	1.5	3
	50	—	1.9	3
1.1	—	120	2	3.5
	120	—	2.5	4
1.5	—	120	2.3	4
	120	—	3	5
2	—	80	3	4.5
	80	220	3.5	5
	220	—	3.8	6
2.1	—	280	4	6.5
	280	—	4.5	7
2.5	—	100	3.8	6
	100	280	4.5	6
	280	—	5	7
3	—	280	5	8
	280	—	5.5	8
4	—	—	6.5	9

❶ Allowable minimum values for the chamfering dimension "r" .

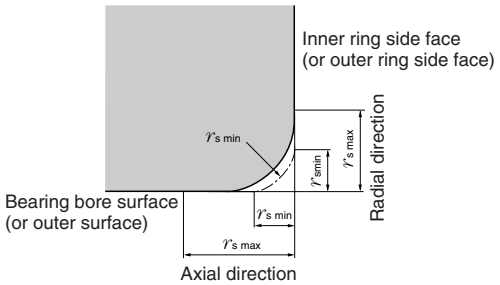


Table 4.5 (2) Thrust bearings

Unit mm

rs min❶	Radial and axial directions rs max
0.3	0.8
0.6	1.5
1	2.2
1.1	2.7
1.5	3.5
2	4
2.1	4.5
3	5.5

❶ Allowable minimum values for the chamfering dimension "r" .

Table 4.6 Basic tolerances

Unit μm

Basic dimension (mm)		IT basic tolerance classes									
over	incl.	IT1	IT2	IT3	IT4	IT5	IT6	IT7	IT8	IT9	IT10
	3	0.8	1.2	2	3	4	6	10	14	25	40
3	6	1	1.5	2.5	4	5	8	12	18	30	48
6	10	1	1.5	2.5	4	6	9	15	22	36	58
10	18	1.2	2	3	5	8	11	18	27	43	70
18	30	1.5	2.5	4	6	9	13	21	33	52	84
30	50	1.5	2.5	4	7	11	16	25	39	62	100
50	80	2	3	5	8	13	19	30	46	74	120
80	120	2.5	4	6	10	15	22	35	54	87	140
120	180	3.5	5	8	12	18	25	40	63	100	160
180	250	4.5	7	10	14	20	29	46	72	115	185
250	315	6	8	12	16	23	32	52	81	130	210
315	400	7	9	13	18	25	36	57	89	140	230
400	500	8	10	15	20	27	40	63	97	155	250
500	630	9	11	16	22	30	44	70	110	175	280
630	800	10	13	18	25	35	50	80	125	200	320
800	1 000	11	15	21	29	40	56	90	140	230	360
1 000	1 250	13	18	24	34	46	66	105	165	260	420
1 250	1 600	15	21	29	40	54	78	125	195	310	500
1 600	2 000	18	25	35	48	65	92	150	230	370	600
2 000	2 500	22	30	41	57	77	110	175	280	440	700
2 500	3 150	26	36	50	69	93	135	210	330	540	860

5. Bearing Internal Clearance

5.1 Radial clearance

Radial clearance in bearing (initial radial clearance) means the displacement value of bearing ring when the non-fixed ring was displaced in radial direction with either one of the inner ring and outer ring remained fixed, before the bearing is installed on a shaft or in a housing.

Machined ring needle roller bearing (with inner ring) ;

The initial clearance values for this bearing type are as shown in **Table 5.1**. **Table 5.1 (1)** shows the interchangeable clearances, which are remained unchanged even after inner ring or outer ring was reassembled in. **Table 5.1(2)** shows non-interchangeable clearances, which do not allow reassembly of inner ring or outer ring due to the narrow clearance range. The bearing clearances are represented by the symbols of C2, ordinary, C3 and C4 in the order from the smallest one and the non-interchangeable clearance symbols are followed by "NA" for identification.

For the radial clearance values for bearing types other than the machined ring needle roller bearings, refer to "Commentary" described in the respective Dimensions Table.

5.2 Running clearance

5.2.1 Setting up running clearance

Bearing clearance in running, that is, running clearance gets generally smaller than the initial radial clearance, depending on the fits and temperature difference between the inner ring and the outer ring. This running clearance must be set up optimally because setting up it improperly would lead to shorter life, overheat and vibration or running noise of bearing.

Theoretically the bearing life comes to the maximum when the running clearance of bearing in normal running is slightly negative, but it is difficult to always hold this optimal condition during actual running. If the negative clearance value gets larger due to change of somewhat operating condition, it would cause significant reduction of bearing lifetime and overheat. To avoid such phenomena, generally proper initial radial clearance is selected and set up so the running clearance gets slightly larger than 0 (zero).

In the case of usual operating conditions, in other words, application of the fits based on ordinary load, usual rotational speed and running temperature, etc., an optimal running clearance can be got by selecting an ordinary clearance.

Table 5.1 Radial clearance in machined ring needle roller bearings

Table 5.1 (1) Interchangeable bearings

Unit μm

Nominal bore diameter d (mm)		Radial clearance							
		C2		Normal ❶		C3		C4	
over	incl.	min	max	min	max	min	max	min	max
—	10	0	30	10	40	25	55	35	65
10	18	0	30	10	40	25	55	35	65
18	24	0	30	10	40	25	55	35	65
24	30	0	30	10	45	30	65	40	70
30	40	0	35	15	50	35	70	45	80
40	50	5	40	20	55	40	75	55	90
50	65	5	45	20	65	45	90	65	105
65	80	5	55	25	75	55	105	75	125
80	100	10	60	30	80	65	115	90	140
100	120	10	65	35	90	80	135	105	160
120	140	10	75	40	105	90	155	115	180
140	160	15	80	50	115	100	165	130	195
160	180	20	85	60	125	110	175	150	215
180	200	25	95	65	135	125	195	165	235
200	225	30	105	75	150	140	215	180	255
225	250	40	115	90	165	155	230	205	280
250	280	45	125	100	180	175	255	230	310
280	315	50	135	110	195	195	280	255	340
315	355	55	145	125	215	215	305	280	370
355	400	65	160	140	235	245	340	320	415
400	450	70	190	155	275	270	390	355	465

❶ Supplementary suffix codes of clearance is not added to bearing numbers.

Table 5.1 (2) Non-interchangeable bearings

Unit μm

Nominal bore diameter d (mm)		Radial clearance							
		C2NA		NA ❷		C3NA		C4NA	
over	incl.	min	max	min	max	min	max	min	max
—	10	10	20	20	30	35	45	45	55
10	18	10	20	20	30	35	45	45	55
18	24	10	20	20	30	35	45	45	55
24	30	10	25	25	35	40	50	50	60
30	40	12	25	25	40	45	55	55	70
40	50	15	30	30	45	50	65	65	80
50	65	15	35	35	50	55	75	75	90
65	80	20	40	40	60	70	90	90	110
80	100	25	45	45	70	80	105	105	125
100	120	25	50	50	80	95	120	120	145
120	140	30	60	60	90	105	135	135	160
140	60	35	65	65	100	115	150	150	180
160	180	35	75	75	110	125	165	165	200
180	200	40	80	80	120	140	180	180	220
200	225	45	90	90	135	155	200	200	240
225	250	50	100	100	150	170	215	215	265
250	280	55	110	110	165	185	240	240	295
280	315	60	120	120	180	205	265	265	325
315	355	65	135	135	200	225	295	295	360
355	400	75	150	150	225	255	330	330	405
400	450	85	170	170	255	285	370	370	455

❷ For bearing with normal clearance, only NA is added to bearing numbers.
EX. NA4920NA

5.2.2 Calculation of running clearance

The running clearance in bearing can be determined from initial radial clearance, reduced internal clearance incurred by effective interference, and temperature difference between inner ring and outer ring, using **formula (5.1)**.

$$\delta_{eff} = \delta_o - (\delta_f + \delta_t) \quad \cdots\cdots\cdots\cdots\cdots\cdots (5.1)$$

where,

δ_{eff} : Running clearance mm
δ_o : Initial radial clearance mm
δ_f : Reduced internal clearance incurred by effective interference mm
δ_t : Reduced internal clearance incurred by temperature difference between inner ring and outer ring mm

(1) Reduced internal clearance incurred by effective interference

When a bearing is installed on a shaft or in a housing with some effective interference, the inner ring expands and the outer ring shrinks and, as the result, the radial clearance in the bearing reduces correspondingly.

The expansion or shrinkage of inner ring or outer ring is approximately equivalent to 85% of the effective interference though depending on bearing type, shaft or housing profile, dimension and material. For the detail refer to **Table 6.4** on page A-35.

$$\delta_f = 0.85 \cdot \Delta_{deff} \quad \cdots\cdots\cdots\cdots\cdots\cdots (5.2)$$

where,

δ_f : Reduced internal clearance incurred by effective interference mm
Δ_{deff} : Effective interference mm

(2) Reduced internal clearance incurred by temperature difference between inner ring and outer ring

When a bearing is in running, the temperature of its outer ring is lower by 5 to 10˚C than that of its inner ring or rolling elements. Also, temperature difference between the inner ring and the outer ring gets larger when heat is radiated more from the housing or when the bearing shaft is communicated with a heat source and a heated fluid is flowing through a hollowed shaft. In such a case, the internal clearance reduces corresponding to thermal expansion difference between the inner ring and the outer ring which is incurred by this temperature difference.

$$\delta_t = \alpha \cdot \Delta_T \cdot D_o \quad \cdots\cdots\cdots\cdots\cdots\cdots (5.3)$$

where,

δ_t : Reduced clearance value incurred by temperature difference mm
α : Linear expansion coefficient of bearing steel 12.5×10^{-6}/˚C
Δ_T : Inner ring – outer ring temperature difference ˚C
D_o : Outer ring raceway diameter mm
d : Bearing bore diameter mm
D : Bearing outer diameter mm

When a shaft or a housing is used as a direct raceway, temperature difference between the shaft and the housing is applied as temperature difference (Δ_T).

5.3 Fits and bearing radial clearance

Where the allowable tolerances for the shaft and the housing hole are already decided, the simple nomogram as shown in **Fig. 5.1** is available as a guideline to decision of the initial radial clearance of bearing so as to enable to get an optimal clearance after the bearing was installed on the shaft/in the housing. The nomogram in **Fig. 5.1** is used as the guideline as stated above. For the detail feel free to contact **NTN**.

For example, where the fit condition for needle roller bearing with inner ring is already given as J7m6, **Fig.5.1** shows that clearance C3 must be secured to get the standard running clearance after installation.

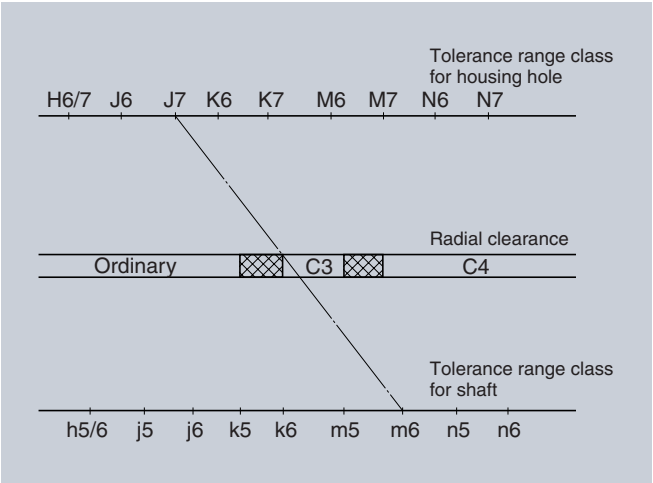


Fig. 5.1 Relationship between bearing fits and radial clearance

7. Bearing Fits

6.1 Interference

For rolling bearings, the inner ring and outer ring are fixed on the shaft or in the housing so that relative movement does not occur between the fitted surfaces of the bearing ring and the shaft or housing in radial, axial and rotational directions when a load acts on the bearing. Such a relative movement, if occurred, would result in wear, fretting corrosion, friction crack, etc. on the fitted surfaces, which would then cause damage of the bearing and the shaft or the housing. Furthermore, wear powder invades into the bearing, then causing imperfect rotation, abnormal overheat, vibration, etc.

The most effective way to fix a bearing is to assign an proper interference to the fitted surfaces between the bearing ring and the shaft or the housing and to thereby apply "tight fit" to the bearing. Furthermore, as its advantage this tight fit method supports the thin-walled bearing ring with uniform load throughout its entire circumference without any loss of load carrying capacity.

The needle roller bearing is a bearing type enabling to separate the inner ring and the outer ring from one another and, therefore, it can be installed on a shaft or in a housing with an interference applied to both of its inner ring and outer ring. In the case of "tight fit", the easiness of bearing installation and removal is lost and, therefore, the bearing ring subjected to stationary load can be "loose-fitted".

6.2 Necessity of proper fit

Improper fit could lead to damage and shorter life of bearing. Therefore, advance careful analysis is needed for selection of proper fit. Representative examples of bearing defects caused by improper fit are as described below.

- Crack and earlier separation of bearing ring, and displacement of bearing ring
- Wear of bearing ring, shaft and housing caused by creep and fretting corrosion
- Seizure (sticking) caused by less internal clearance
- Insufficient running accuracy and abnormal noise caused by deformed raceway surface

6.3 Fit selection

Fit selection is generally done in accordance with the rule specified hereunder.

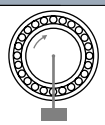
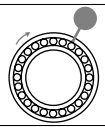
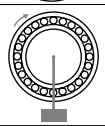
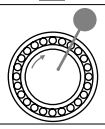
The loads acting on each bearing ring are divided into running load, stationary load and directionally unstable load according to the direction and characteristic of loads acting on the bearing.

"Tight fit" can be selected for a bearing ring subjected to running load and directionally unstable load and "stationary fit" or "loose fit" be selected for a bearing ring subjected to stationary load. (See **Table 6.1**)

Where load of high magnitude or vibration and shock loads act on a bearing or a light alloy/plastic housing is used, it is necessary to secure a large interference. In that case, **however, the housing rigidity must be considered carefully so as not allow occurrence of split damage, etc.**

For an application subjected to high running accuracy, bearings of high accuracy must be used with a shaft and a housing of higher dimensional accuracy so as not to apply a large interference thereto. Applying a large interference would cause the shaft or housing profile to be transferred to the bearing track, which could then interfere with the bearing running accuracy. The said bearings of high accuracy are used to prevent such possible phenomena.

Table 6.1 Radial load and bearing fit

Bearing running conditions	Sketch	Load characteristic	Bearing fit	
			Inner ring	Outer ring
Inner ring : Rotation Outer ring: static Load direction: constant		Rotating inner ring load	Tight fit	Loose fit acceptable, too
Inner ring: static Outer ring: rotation Load direction: rotating with outer ring		Static outer ring load		
Inner ring: static Outer ring: rotation Load direction: constant		Rotating inner ring load	Loose fit acceptable, too	Tight fit
Inner ring: rotation Outer ring: static Load direction : rotating with inner ring		Static outer ring load		
Inner ring: rotation or static Outer ring: rotation or static Load direction: The direction can not be fixed.	Load direction is non-constant due to directional fluctuation, unbalanced load, etc.	Directionally unstable load	Tight fit	Tight fit

6.4 Recommended Fits

The dimensional tolerances for the diameter of a shaft and the hole diameter of a bearing housing, on/in which a bearing is installed, are standardized under the metric system in ISO 286 and JIS B 0401 (Dimensional Tolerances and Bearing Fits). Hence, bearing fits is determined by selection of the dimensional tolerances for shaft diameter and hole diameter as applicable. **Fig. 6.1** shows the relationship between shaft diameter and bearing bore diameter and between housing hole diameter and bearing outer diameter.

Table 6.2 shows the recommended fits for the machined ring needle roller bearings (with inner ring) that are generally selected based on the dimensional and load conditions. **Table 6.3** shows the numerical fit values.

For the recommended fits for others than the machined ring needle roller bearings, refer to "Commentary" described in the respective Dimension Tables.

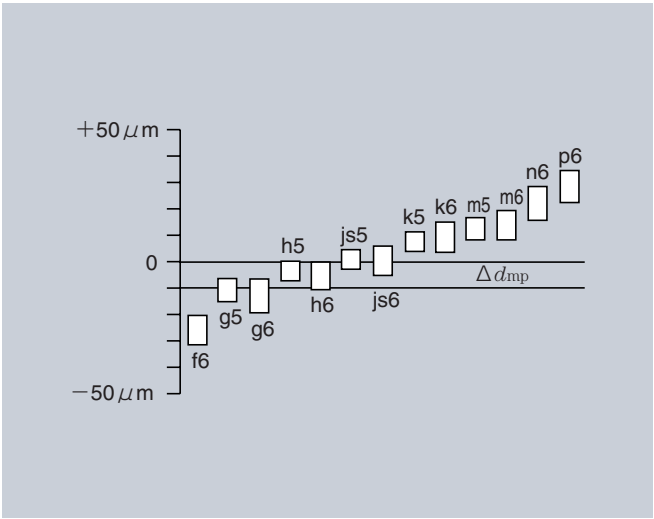


Fig. 6.1 (a) Bearing fits on shaft

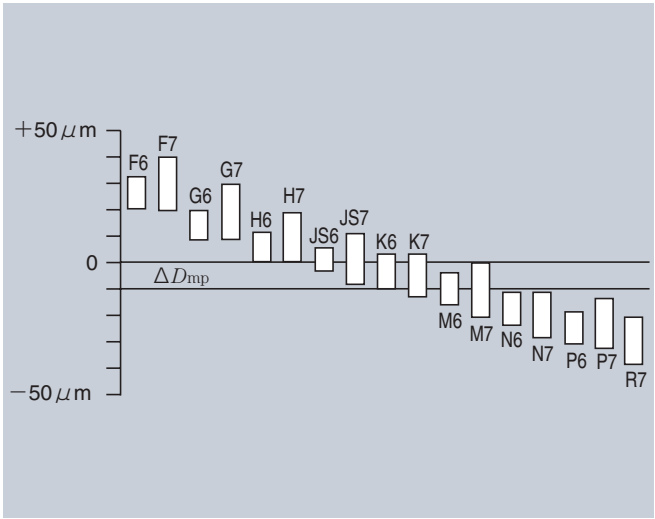


Fig.6.1(b) Bearing fits in housing hole

Table 6.2 Machined ring needle roller bearing fits
Table 6.2 (1) Dimensional tolerance for shaft diameter

Conditions			Tolerance range class
Load characteristic	Load magnitude	Shaft diameter d mm	
Rotating inner ring load or directionally unstable load	Light load	～ 50	j5
	Ordinary load	～ 50	k5
		50～150	m5
		150～	m6
	Heavy load and shock load	～150	m6
		150～	n6
Inner ring static load	Medium- and low-speed rotation, light load	All dimensions	g6
	General application		h6
	When high rotational accuracy is required		h5

Table 6.2 (2) Dimensional tolerances for housing

Conditions		Tolerance range class
Outer ring static load	Ordinary and heavy load	J7
	Two-split housing, ordinary load	H7
Rotating outer ring load	Light load	M7
	Ordinary load	N7
	Heavy load and shock load	P7
Directionally unstable load	Light load	J7
	Ordinary load	K7
	Heavy load and shock load	M7
When high rotational accuracy under light load is required		K6

Remarks: Light load, ordinary load and heavy load are classified per the following criteria.
Light load : $Pr \leqq 0.06Cr$
Ordinary load : $0.06Cr < Pr \leqq 0.12Cr$
Heavy load : $Pr > 0.12Cr$

Table 6.3 Numerical fit values for radial bearing (JIS Class-0)
Table 6.3(1) Bearing fits on shaft

Unit μm

Nominal bore diameter d mm over	incl.	Allowable deviation of mean bore diameter Δ_{dmp} high	low	g6 Bearing	g6 Shaft	h5 Bearing	h5 Shaft	h6 Bearing	h6 Shaft	j5 Bearing	j5 Shaft	k5 Bearing	k5 Shaft	m5 Bearing	m5 Shaft	m6 Bearing	m6 Shaft	n6 Bearing	n6 Shaft
3	6	0	−8	4T～12L		8T～ 5L		8T～ 8L		11T～ 2L		14T～1T		17T～ 4T		20T～ 4T		24T～ 8T	
6	10	0	−8	3T～14L		8T～ 6L		8T～ 9L		12T～ 2L		15T～1T		20T～ 6T		23T～ 6T		27T～10T	
10	18	0	−8	2T～17L		8T～ 8L		8T～11L		13T～ 3L		17T～1T		23T～ 7T		26T～ 7T		31T～12T	
18	30	0	−10	3T～20L		10T～ 9L		10T～13L		15T～ 4L		21T～2T		27T～ 8T		31T～ 8T		38T～15T	
30	50	0	−12	3T～25L		12T～11L		12T～16L		18T～ 5L		25T～2T		32T～ 9T		37T～ 9T		45T～17T	
50	80	0	−15	5T～29L		15T～13L		15T～19L		21T～ 7L		30T～2T		39T～11T		45T～11T		54T～20T	
80	120	0	−20	8T～34L		20T～15L		20T～22L		26T～ 9L		38T～3T		48T～13T		55T～13T		65T～23T	
120	140	0	−25	11T～39L		25T～18L		25T～25L		32T～11L		46T～3T		58T～15T		65T～15T		77T～27T	
140	160																		
160	180																		
180	200	0	−30	15T～44L		30T～20L		30T～29L		37T～13L		54T～4T		67T～17T		76T～17T		90T～31T	
200	225																		
225	250																		
250	280	0	−35	18T～49L		35T～23L		35T～32L		42T～16L		62T～4T		78T～20T		87T～20T		101T～34T	
280	315																		
315	355	0	−40	22T～54L		40T～25L		40T～36L		47T～18L		69T～4T		86T～21T		97T～21T		113T～37T	
355	400																		
400	450	0	−45	25T～60L		45T～27L		45T～40L		52T～20L		77T～5T		95T～23T		108T～23T		125T～40T	
450	500																		

Table 6.3 (2) Bearing fits in housing hole

Unit μm

Nominal outer diameter D mm over	incl.	Allowable deviation of mean outer diameter Δ_{Dmp} high	low	H7 Housing	H7 Bearing	J7 Housing	J7 Bearing	K6 Housing	K6 Bearing	K7 Housing	K7 Bearing	M7 Housing	M7 Bearing	N7 Housing	N7 Bearing	P7 Housing	P7 Bearing
6	10	0	−8	0～ 23L		7T～16L		7T～10L		10T～13L		15T～ 8L		19T～ 4L		24T～ 1L	
10	18	0	−8	0～ 26L		8T～18L		9T～10L		12T～14L		18T～ 8L		23T～ 3L		29T～ 3L	
18	30	0	−9	0～ 30L		9T～21L		11T～11L		15T～15L		21T～ 9L		28T～ 2L		35T～ 5L	
30	50	0	−11	0～ 36L		11T～25L		13T～14L		18T～18L		25T～11L		33T～ 3L		42T～ 6L	
50	80	0	−13	0～ 43L		12T～31L		15T～17L		21T～22L		30T～13L		39T～ 4L		52T～ 8L	
80	120	0	−15	0～ 50L		13T～37L		18T～19L		25T～25L		35T～15L		45T～ 5L		59T～ 9L	
120	150	0	−18	0～ 58L		14T～44L		21T～22L		28T～30L		40T～18L		52T～ 6L		68T～10L	
150	180	0	−25	0～ 65L		14T～51L		21T～29L		28T～37L		40T～25L		52T～13L		68T～ 3L	
180	250	0	−30	0～ 76L		16T～60L		24T～35L		33T～43L		46T～30L		60T～16L		79T～ 3L	
250	315	0	−35	0～ 87L		16T～71L		27T～40L		36T～51L		52T～35L		66T～21L		88T～ 1L	
315	400	0	−40	0～ 97L		18T～79L		29T～47L		40T～57L		57T～40L		73T～24L		98T～ 1L	
400	500	0	−45	0～108L		20T～88L		32T～53L		45T～63L		63T～45L		80T～28L		108T～ 0	

Remarks: Fit symbols “L” and “T” represent bearing clearance and interference respectively.

6.5 Interference Calculation

(1) Load and required interference

When radial load acts on a bearing, the interference required to prevent a clearance between its inner ring and a steel solid shaft can be expressed in **formulas (6.1)** and **(6.2)**.

For $F_r \leqq 0.3\ C_{or}$,

$$\Delta d_F = 0.08\sqrt{\frac{d \cdot F_r}{B}} \left(0.25\sqrt{\frac{d \cdot F_r}{B}}\right) \cdots\cdots\cdots\cdots (6.1)$$

For $F_r > 0.3\ C_{or}$,

$$\Delta d_F = 0.02\frac{F_r}{B} \left(0.2\frac{F_r}{B}\right) \cdots\cdots\cdots\cdots (6.2)$$

Where,

Δd_F : Required effective interference mm
d : Bearing bore diameter mm
B : Inner ring width mm
F_r : Radial load N (kgf)
C_{0r} : Basic static load rating N (kgf)

(2) Temperature rise and required interference

When temperature rise of bearing (difference between bearing temperature and ambient temperature) is inevitable incurred by bearing running, the interference required to prevent a clearance between the inner ring and a steel shaft can be expressed in **formula (6.3)**.

$$\Delta d_F = 0.0015 \cdot d \cdot \Delta T \cdots\cdots\cdots\cdots (6.3)$$

Where,

Δd_T : Required effective interference for temperature difference μm
ΔT : Difference between bearing temperature and ambient temperature ˚C
d : Bearing bore diameter mm

(3) Fitting surface roughness and required interference

The fitting surface is smoothed (surface roughness is made less) by bearing fits so that the interference reduces correspondingly. The interference reduced value differs depending on the fitting surface roughness and generally the following reduction values must be prospected.

For ground shafts : 1.0 to 2.5mm
For lathe-turned shafts : 5.0 to 7.0 mm

(4) Maximum interference

Bearing ring fitted, with interference, on a shaft or in a housing results in tensile stress or compressive stress. Over-interference could cause cracking /splitting of bearing and short fatigue life of bearing. Therefore, in general the maximum interference is secured at 1/1000 and less of shaft diameter or otherwise it is secured so the circumferential maximum stress generating on the fitting surface comes to 130MPa or less. (See **Table 6.4**)

(5) Stress and deformation caused by interference

When bearing ring (solid) is fitted with interference, it deforms elastically and this elastic deformation results in stress.(See **Fig.6.2**) The fitting surface pressure of bearing ring, circumferential tensile stress (inner ring), compressive stress (outer ring) and radial expansion of raceway (inner ring), and shrinkage(outer ring) can be calculated from **Table 6.4**.

Table 6.4 Deformation and stress caused by bearing fit

Item	Inner ring	Outer ring
Surface pressure p MPa	$p_i = \frac{E}{2}\frac{\Delta d_{eff}}{d}\frac{(1-k^2)(1-k_0^2)}{1-k^2k_0^2}$	$p_e = \frac{E}{2}\frac{\Delta D_{eff}}{D}\frac{(1-h^2)(1-h_0^2)}{1-h^2h_0^2}$
Circumferential maximum stress σ MPa	$\sigma_i = p_i\frac{1+k^2}{1-k^2}$ (Tensile stress)	$\sigma_e = p_e\frac{2}{1-h^2}$ (Compressive stress)
Radial elastic deformation of raceway Δ	$\Delta_i = \Delta d_{eff} \cdot k\frac{1-k_0^2}{1-k^2k_0^2}$ (Expansion)	$\Delta_e = \Delta D_{eff} \cdot h\frac{1-h_0^2}{1-h^2h_0^2}$ (Shrinkage)

Where,

$k = \frac{d}{d_i}$, $k_0 = \frac{d_0}{d}$, $h = \frac{D_e}{D}$, $h_0 = \frac{D}{D_0}$

Remarks (Symbol representation)

d : Inner ring bore diameter (shaft diameter) mm
d_0 : Hollowed shaft bore diameter (For solid shaft, $d_0 = 0$) mm
d_i : Inner ring raceway diameter mm
Δd_{eff} : Effective interference for inner ring mm
D : Outer ring outer diameter (housing hole diameter) mm
D_0 : Housing outer diameter (For sufficient housing size, $D_0 = \infty$) mm
D_e : Outer ring raceway diameter mm
ΔD_{eff} : Effective interference for outer ring mm
E : Modulus of elasticity (Young factor) 2.07×10^6 (21200) MPa (kgf/mm^2)

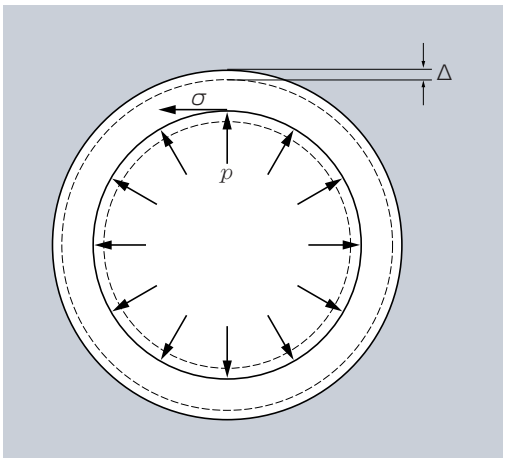


Fig.6.2

7. Shaft and Housing Design

Even if the bearing to be used is selected correctly, it can not fulfill its specific function unless the shaft/housing on/in which it is installed is designed correctly. Particularly for needle roller bearings the shaft and the housing must be designed under special consideration because the bearing ring thickness is thinner than that of other rolling bearings.

7.1 Design of bearing installing portions

Where a needle roller bearing with cage is used as an individual and is guided in axial direction directly by shaft shoulder (**Fig.7.1**), the shaft shoulder with which the cage side face comes in contact must be finished accurately and also is not allowed to have any raiser thereon. For a needle roller bearing subjected to high speed running, its contact surface is hardened and thereafter fine-finished by grinding.

When a snap ring is used for axial guide of the cage (**Fig.7.1**), a thrust ring is inserted between the cage and the snap ring to prevent the cut section of the snap ring from contacting directly the cage. **NTN** shaft snap ring WR type specially designed for **NTN** needle roller bearings is available as the snap ring intended for axial guide of bearing cage. (Refer to Dimensions Table on page B-227.)

On other hand, radial needle roller bearing can move freely in axial direction so that a ball bearing or a thrust bearing is used at single side for axial positioning of the shaft. Where axial load is less and, in addition, rotational speed is not so high(Ex. idle gear in a gear box), a thrust ring is fitted on the shaft, as illustrated in **Fig. 7.2** and a sliding bearing is formed between the thrust ring and the outer ring or the housing end face for the purpose of axial positioning. In such a design, good care must be exercised so the guide surface is fully lubricated. **Fig.7.3** illustrates an example of the above thrust ring with oil groove on its guide surface. The boundary between this oil groove and the plane area must be chamfered for deburring.

In general, for proper installation of needle roller bearing the inner ring and outer ring are both positioned in axial direction so that the bearing displaces in axial direction while running.

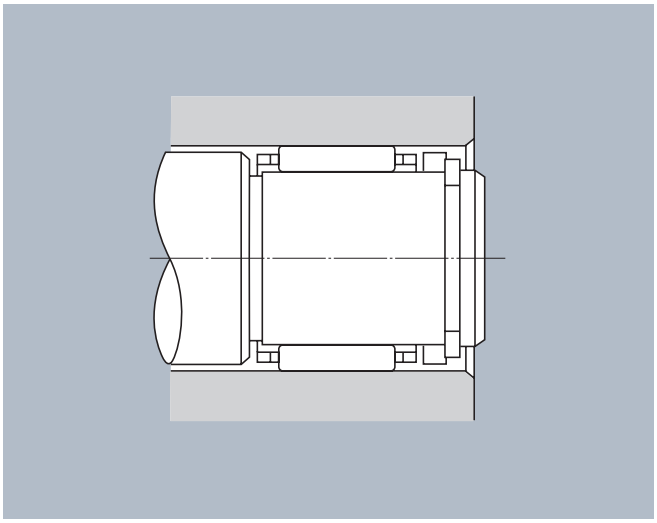


Fig. 7.1 Bearing fixing by thrust ring

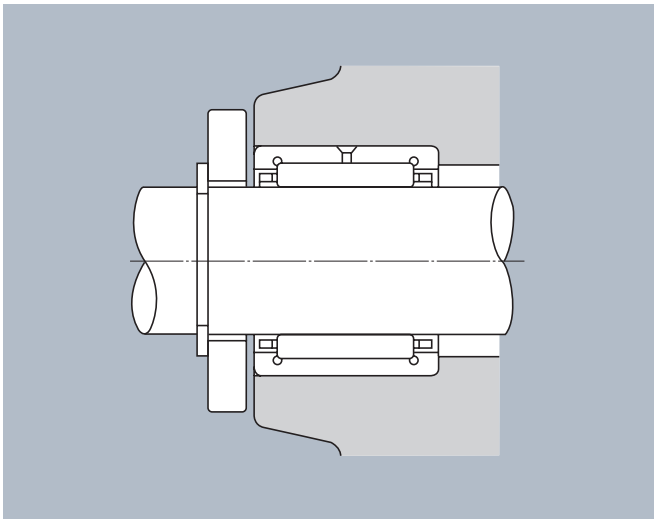


Fig. 7.2 Bearing fixing in axial direction

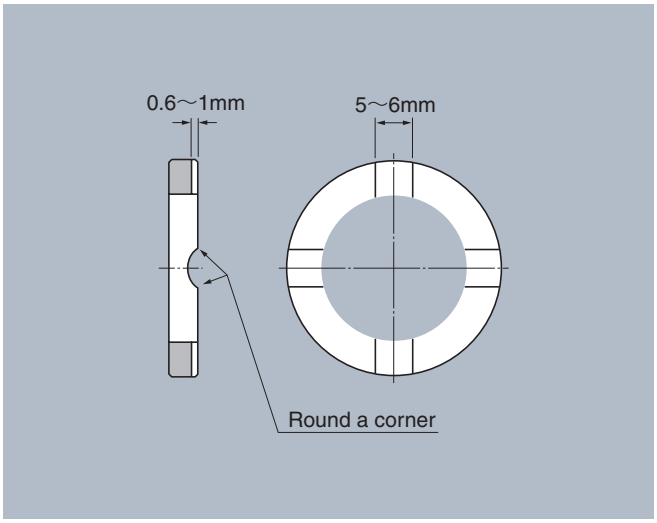


Fig. 7.3 Design of thrust ring guide surface

(1) Inner ring

For fixing inner ring correctly on a shaft, the shaft shoulder face is finished at the right angle against the shaft axial center and, in addition, the shaft corner is rounded smaller than the inner ring chamfer dimension.

Where the corner roundness $r_{a\ max}$ must be made larger for specific shaft strength (**Fig.7.4a**) or the shaft shoulder is low (**Fig. 7.4b**), a spacer must be inserted between the shaft shoulder and the inner ring to hold a full contact surface with the inner ring. Furthermore, for facilitating inner ring pull-out work the shaft shoulder is provided with a notched groove,as illustrated in **Fig. 7.5**, to accept an inner ring pull-out jig (puller).

NTN snap ring WR type for shaft use (Refer to Dimensions Table on page B-227) can be used for simply fixing inner ring in axial direction. (**Fig. 7.6**) Moreover, inner ring can be fixed in axial direction using an end plate or a side ring as illustrated in **Figs. 7.7** and **7.8**.

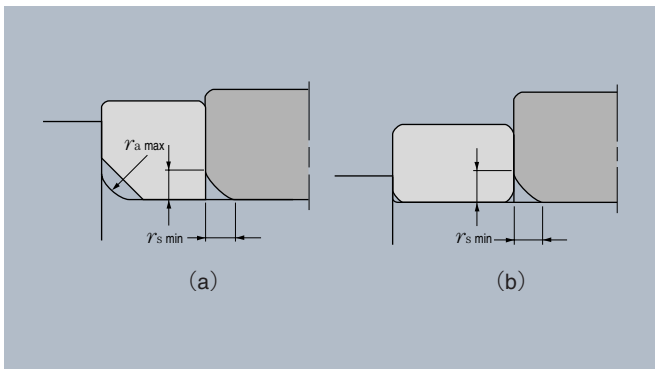


Fig. 7.4 Inner ring fixing by use of spacer

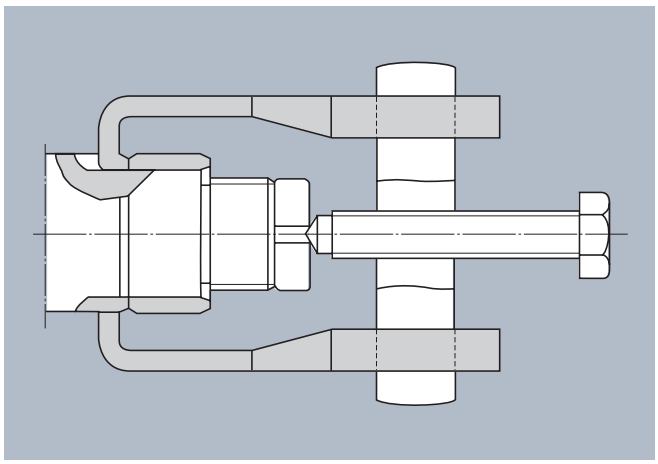


Fig. 7.5 Inner ring pull-out jig(puller)

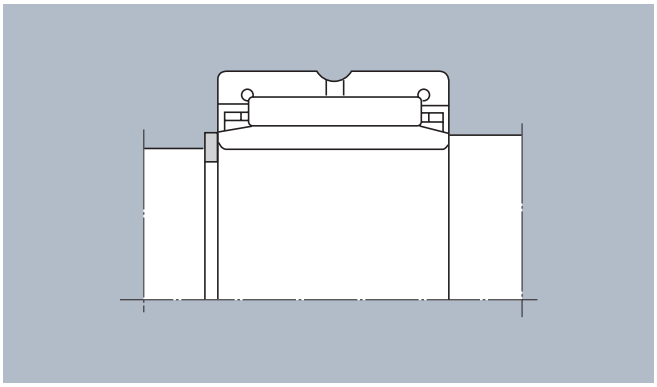


Fig. 7.6 Inner ring fixing by snap ring

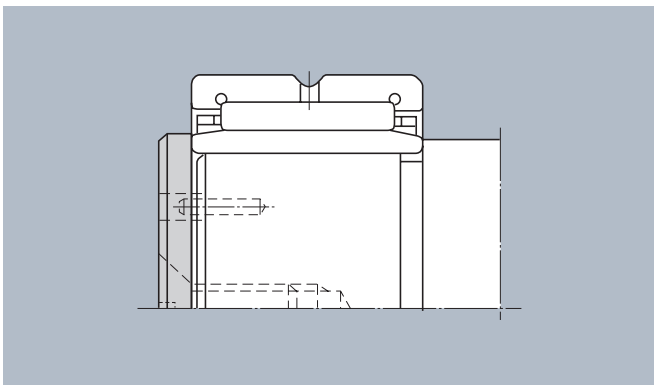


Fig. 7.7 Inner ring fixing by end plate

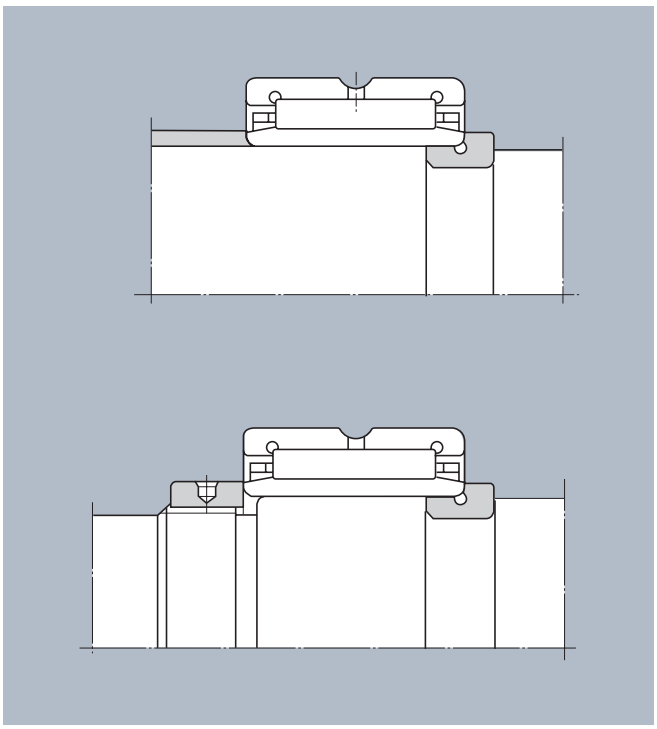


Fig.7.8 Inner ring fixing by side ring

(2) Outer ring

Similarly to Para. 7.1(1) "Inner Ring", good care exercised of the shoulder profile of bearing housing for fixing outer ring in axial direction.

Figs. 7.9 and **7.10** illustrate the methods of fixing outer ring in axial direction.

NTN housing snap ring BR type (Refer to Dimensions Table on page B-229) can be used for fixing outer ring in axial direction. **NTN** BR type snap rings are designed to the dimensions adaptable to the needle roller bearings with low section height. However, commercially available snap rings conforming to JIS standard as applicable can also be used for the same bearings with adequately high section height.

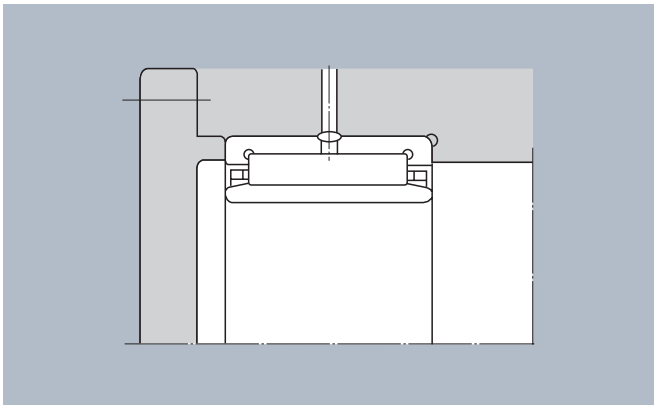


Fig. 7.9 Outer ring fixing by lid

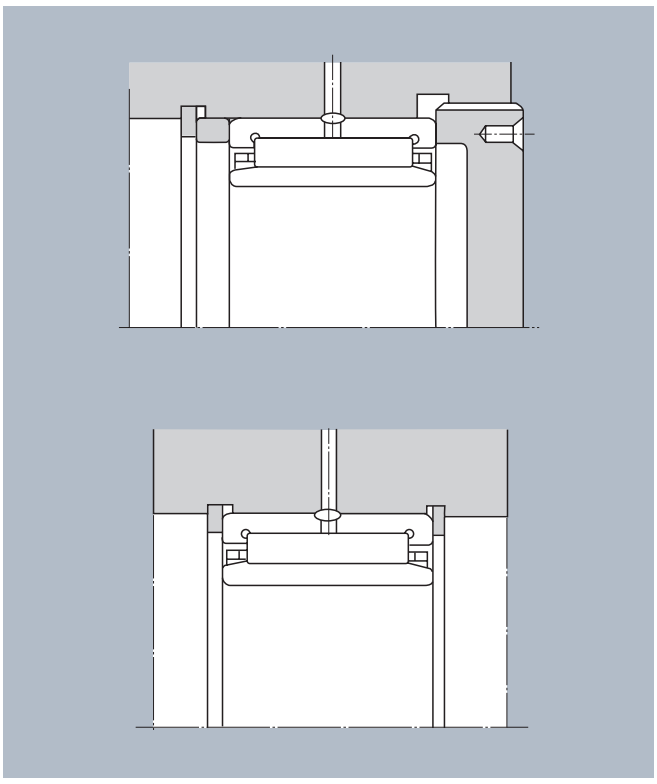


Fig. 7.10 Outer ring fixing by snap ring

7.2 Bearing fitting dimensions

The respective shoulder heights "h" of shaft and housing are designed larger than the maximum chamfer dimension $r_{s\ max}$ for bearing so the bearing end face comes in contact with the flat zone. And the corner roundness "r_{as}" is designed smaller than the minimum chamfer dimension "r_s" of bearing so as not to interfere with the bearing. Generally the radius of shaft and housing corner roundness shown in **Table 7.1** is used as the shoulder heights of shaft and housing.

The dimensions of the shafts and housings related to bearing installation are as described in the dimensions table for each bearing type. The shoulder diameter shown in this table means the effective shoulder diameter which comes in contact with the side face of bearing excluding the chamfered portion of shoulder.

When the fitting surface of shaft or housing is finished by grinding, the shoulder corner must be provided with a relief. This relief dimension is as shown in **Table 7.2**.

Table 7.1 Radius of shaft /housing corner roundness and shoulder height

Unit mm

$r_{s\ min}$	$r_{as\ max}$	h (min)
0.15	0.15	0.6
0.2	0.2	0.8
0.3	0.3	1
0.6	0.6	2
1	1	2.5
1.1	1	3.25
1.5	1.5	4
2	2	4.5
2.1	2	5.5
2.5	2	6
3	2.5	6.5
4	3	8

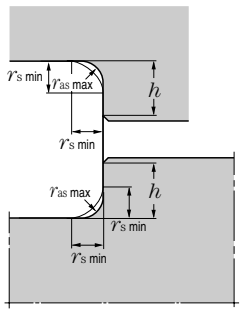
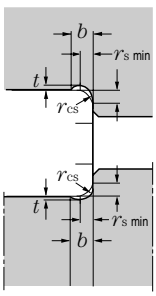


Table 7.2 Relief grinding dimension for shaft and housing corners

Unit mm

$r_{s\ min}$	b	t	r_{cs}
1	2	0.2	1.3
1.1	2.4	0.3	1.5
1.5	3.2	0.4	2
2	4	0.5	2.5
2.1	4	0.5	2.5
3	3.7	0.5	3
4	5.9	0.5	4



7.3 Shaft and housing accuracy

Because of its thin wall, the raceway surface accuracy of needle roller bearing ring is influenced by the respective fitting surface accuracy of a shaft /a housing on/in which the needle roller bearing is installed. In the case of general operating conditions, the fitting surfaces of shaft and housing may be finished by lathe-turning, but those must be finished by grinding where the acting load is great and, in addition, the requirements for accuracy and sound are strict.

Table 7.3 shows the fitting surface accuracy, profile accuracy, surface roughness, and shoulder perpendicularity to fitting surface of each of shaft and housing under ordinary operating conditions.

Table 7.3 Shaft and housing accuracy

Characteristic item	Shaft	Housing
Dimensional accuracy	IT6（IT5）	IT7（IT6）
Roundness cylindricality (max)	IT3（IT2）	IT4（IT3）
Shoulder perpendicularity (max)	IT5（IT4）	IT5（IT4）
Fitting surface roughness	0.8a	1.6a

Remarks: The parenthesized values are applied to the bearings of accuracy class 5 and higher.

Where a two-split housing is used, deformation of outer ring by housing clamping can be minimized by providing a relief on the mating surface at bore side.

7.4 Raceway surface accuracy

For needle roller bearings, shaft and housing are used as the raceway surface on application. The raceway dimensional accuracy, profile accuracy and surface roughness of shaft/housing must be equivalent to the raceway accuracy of bearing itself. **Table 7.4** shows the specified surface accuracy and surface roughness of shaft/housing raceway.

Table 7.4 Raceway surface accuracy (recommendation)

Characteristic item	Shaft	Housing
Dimensional accuracy	IT5（IT4）	IT6（IT5）
Roundness cylindricality (max)	IT3（IT2）	IT4（IT3）
Shoulder perpendicularity (max)	IT3（IT2）	IT3（IT2）
Axial run-out Thrust bearing (max)	IT5（IT4）	
Surface roughness	For shaft diameter of ϕ 80 and less :0.2a For shaft diameter of over ϕ 80 to 120 :0.3a For shaft diameter of over ϕ 120 :0.4a	

Note) The parenthesized values are applied where high rotational accuracy is required.

7.5 Material and hardness of raceway surface

When the outer surface or bore surface of shaft(hollowed) or housing is used as raceway, it must be hardened to HRC58 to 64 for getting sufficient load capacity. For that, the materials shown in **Table 7.5** are used after heat-treated properly.

Table 7.5 Materials used for raceway

Kinds of steel	Representative example	Related standards
High carbon chrome bearing steel	SUJ2	JIS G 4805
Carbon tool steel	SK3	JIS G 4401
Nickel chrome molybdenum steel	SNCM420	JIS G 4103
Chrome steel	SCr420	JIS G 4104
Chrome molybdenum steel	SCM420	JIS G 4105
Nickel chrome steel	SNC420	JIS G 4102

When steel is surface-hardened by carburizing or carbonitriding, JIS Standard as applicable defines the depth from surface up to HV550 as an effective hardened layer. The minimum value of effective hardened layer depth is approximately expressed in **formula (7.1)**.

$$E_{ht\ \min} \geqq 0.8 D_w \ (0.1+0.002 D_w) \quad \cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots (7.1)$$

Where,

$E_{ht\ \min}$: Minimum effective hardened layer depth mm

D_w : Roller diameter mm

7.6 Allowable bearing inclination

The inner ring and outer ring of bearing incline a little eventually against one another depending on shaft deflection, shaft /housing machining accuracy, fitting deviation, etc. Although this allowable inclination differs depending on bearing type, bearing load, internal clearance, etc., the inclination degree shown in **Table 7.6** must be used as a guideline in the case of general applications because even minor inclination of inner ring and outer ring could cause reduction of bearing life and damage of cage.

Table 7.6

Bearing type	Allowable inclination
Radial needle roller bearing	1/2 000
Thrust bearing	1/10 000

8. Lubrication

The purpose of lubricating bearings is to form a thin oil film on the rolling and sliding surfaces and to thereby prevent metal to metal direct contact. Furthermore, lubricating rolling bearings has the following effects.

(1) Reduction of friction and wear
(2) Discharge of friction heat
(3) Further extension of bearing life
(4) Rusting prevention
(5) Prevention of foreign matter invasion

For achieving these lubrication effects it is necessary to select a good quality lubricant, to remove dust from the lubricant, and to design a proper seal structure for prevention of lubricant leak as well as to adopt the lubrication method suitable to the respective actual operating conditions.

Rolling bearings are mostly lubricated with grease or oil, but solid lubricants such as molybdenum disulfide, graphite, etc. are used for special application.

Further, the comparative data of grease lubrication and oil lubrication given in **Table 8.1** can be utilized as the selection guide.

Table 8.1 Characteristic comparison of grease lubrication with oil lubrication

Comparative items \ Lubrication method	Grease lubrication	Oil lubrication
Handling	◎	△
Reliability	○	◎
Cooling effect	×	○ (Recirculation needed)
Seal structure	○	△
Power loss	○	○
Environmental pollution	○	△
High speed running of bearing	×	○

◎ : Extraordinarily advantageous ○ : Advantageous
△ : Fairly advantageous × : Disadvantageous

8.1 Grease lubrication

Grease lubrication is the simplest lubrication method. This method enabling to simplify design of the seal structure is broadly used.

Important points in this lubrication method are to select an optimum grease and to fill it securely in a bearing. Particularly where the cage is guided by the inner ring or outer ring of bearing, good care must be exercised so the guide surface is fully greased throughout its entire area.

Furthermore, for grease replenishment it is desirable to provide a grease sector and a grease valve, etc. as a grease replenisher. **Fig. 8.1** illustrates an example of bearing unit with grease sector and grease valve.

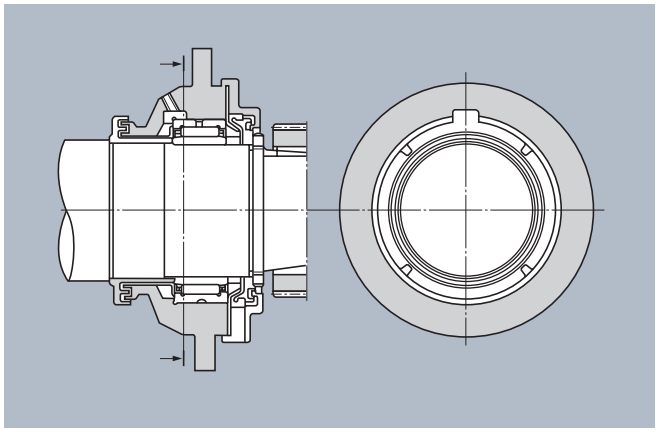


Fig. 8.1 An example of bearing unit with grease sector and grease valve

8.1.1 Types and characteristics of grease

Lubrication oil is composed of a lubrication base oil (ex. mineral oil base or a synthetic oil base) held with a thickener and various additives added thereto. The properties of grease is determined by the kinds and combination of base oil, thickener and additives.

Table 8.2 shows the general types of grease and their characteristics. Even the grease of same type is significantly different in the performance depending on the grease brand selected. In selecting a grease brand, therefore, it is necessary to verify the property data submitted from the grease manufacturer.

8.1.2 Base oil

Mineral oil or synthetic oils such as diester oil, and synthetic oil such as silicone oil, fluorocarbon oil, etc. are used as the base oil for grease.

The lubrication performance of grease is mainly determined by the lubrication performance of the base oil. In general, a grease composed of low viscosity base oil is excellent in low temperature characteristic and high speed performance and, on the other hand, a grease composed of high viscosity base oil is excellent in high load characteristic.

8.1.3 Thickener

Thickener is blended and diffused in base oil to hold grease in semi-solid form. And various thickeners are available for use as follows; thickener composed of metallic soap such as lithium, sodium or calcium, etc., thickener composed of inorganic material such as silica gel, bentonite, etc., and non-soap base thickener composed of organic material such as polyurea, fluorocarbon, etc.

The grease characteristics such as critical operating temperature, mechanical stability, durability, etc. are mainly determined by the kind of thickener used. In general, the sodium soap base grease is inferior to others in water resistance. Non-metallic soap base thickener composed of bentone, polyurea, etc. is excellent in high temperature characteristic.

Table 8.2 Grease varieties and characteristics

Grease name	Lithium grease			Sodium grease (Fiber grease)	Calcium compound base grease	Aluminum grease	Non-soap grease	
Thickener	Li soap			Na soap	Ca+Na soap Ca+Li soap	Al soap	Bentone, silica gel, urea, carbon black, etc.	
Base oil	Mineral oil	Diester oil	Silicone oil	Mineral oil	Mineral oil	Mineral oil	Mineral oil	Synthetic oil
Dropping point ˚C	170～190	170～190	200～250	150～180	150～180	70～90	250 or more	250 or more
Operating temperature range ˚C	−30～+130	−50～+130	−50～+160	−20～+130	−20～+120	−10～+80	−10～+130	−50～+200
Mechanical stability	Excellent	Good	Good	Excellent to good	Excellent to good	Good to poor	Good	Good
Pressure resistance	Good	Good	Poor	Good	Excellent to good	Good	Good	Good
Water resistance	Good	Good	Good	Good to poor	Good to poor	Good	Good	Good
Applications	Broadest application. Grease for universal type rolling bearings.	Excellent in low temperature characteristic and anti-friction characteristic.	suited to high temperature and low temperature. Low oil film strength and unsuitable for high load application.	emulsified by inclusion of water content. Comparatively excellent in high temperature characteristic.	Excellent in water resistance and mechanical stability. Suitable for bearing subjected to shock load.	Excellent in viscosity characteristic. Suitable for bearing subjected to vibration.	Available for use in wide temperature range from low to high temperature. Some of non-soap base greases are excellent in heat resistance, cold resistance, chemical resistance, etc. subject to proper combination of base oil and thickener. Grease for universal type rolling bearings.	

Remarks: The operating temperature range in this table is the general characteristic value, not the guaranteed value.

8.1.4 Additives

Any greases contain various additives to improve the performance, for example, containing oxidation inhibitor, extreme pressure additives (EP additives), rust inhibitor, corrosion inhibitor, etc.

A grease containing extreme pressure additives is used for bearings subjected to high load or shock load. A grease containing oxidation stabilizer is used for bearing applications wherein the operating temperature is comparatively high and no grease is replenished for a long time.

8.1.5 Consistency

"Consistency" is an index showing the hardness or fluidity of grease. The greater numerical value thereof is the softer hardness. This consistency is determined by the amount of thickener and the viscosity of base oil. Usually NLGI consistency codes 1 and 2 or 3 are used for lubrication of bearings.

Table 8.3 shows the general relationship of grease consistency to application.

8.1.6 Grease mixing

When different greases are mixed together, the consistency of the mixed grease varies (generally softens) so that the allowable operating temperature gets lower. To avoid such characteristic variation of grease, **it is not allowed to mix different greases, except mixing greases of same brand.**

Where mixing of different greases is inevitable, greases composed of thickener of same kind and similar base oil must be selected. Even when greases of same kind are mixed together, thus, the properties of the mixed grease could vary depending on difference in additives, etc. It is therefore necessary to check the property variation in advance.

Table 8.3 Grease consistency

NLGI consistency No.		JIS (ASTM) 60-cycle mixed grease consistency	Application
0	Soft	355～385	For centralized greasing
1	⇕	310～340	For centralized greasing
2	⇕	265～295	For general application, for tight-sealed bearing
3	⇕	220～250	For general application, for high temperature
4	Hard	175～205	Special application

8.1.7 Grease fill amount

Grease fill amount differs depending on housing design, spacing volume, rotational speed, kind of grease, etc.

Around 50% to 80% of static spacing volume in bearing and housing is deemed as a guideline to the fill amount. In the case of high rotational speed, this fill amount must be set up a little bit less for controlling temperature rise to a low rate. **Too much fill amount of grease would cause the grease temperature to rise higher, which would then lead to reduction of the specific lubrication performance due to leak of the softened grease or quality change such as oxidation, etc.**

Further, for the machined ring needle roller bearing with inner ring the approximate value of spacing volume in the bearing can be determined by **formula (8.1)**.

$$V = 35W \quad \cdots\cdots (8.1)$$

where,

V : Internal spacing volume (approximate value) cm^3

W : Bearing weight kg (See Dimensions Table)

8.1.8 Grease replenishment

Lubrication grease must be replenished at an proper interval because its lubrication performance deteriorates with elapse of the bearing running time. However, this replenishing interval differs depending on bearing type, dimension, rotational speed, bearing temperature, kind of grease used, etc.

Fig.8.2 gives the replenishing interval chart as a guideline. This chart is subject to use of a grease for ordinary rolling bearings under usual operating conditions.

Needless to say, the grease replenishing interval must be the shorter as the bearing temperature gets higher. As an approximate guideline, when the bearing temperature is 80˚C or more, the replenishing interval shall be shortened by 1/1.5 whenever the bearing temperature rises by 10˚C.

[Ex.] Determine the grease replenishing interval in the case when bearing **NA4910R** is in running under the conditions of radial load P_r 5kN {510kgf} and rotational speed n 1 600r/min.

d =50mm,
n_o =4700 is determined from NA4910R
C_r =48kN (4900kgf) and allowable rotational speed =4,700 r/min in the Dimensions Table.
Accordingly,

$$\frac{n_o}{n} = \frac{4700}{1600} \fallingdotseq 2.9$$

Plot a line horizontally from d =50 point in **Fig. 8.2** and deem the intersection point with the vertical line Ⅰ as A. Thereafter, connect n_o/n=3 point B on the vertical line Ⅱ and said A point together, with a straight line, and determine the intersection point C with the vertical line Ⅲ. And the grease replenishing interval of approximately 4600 hours can be read from the intersection point C.

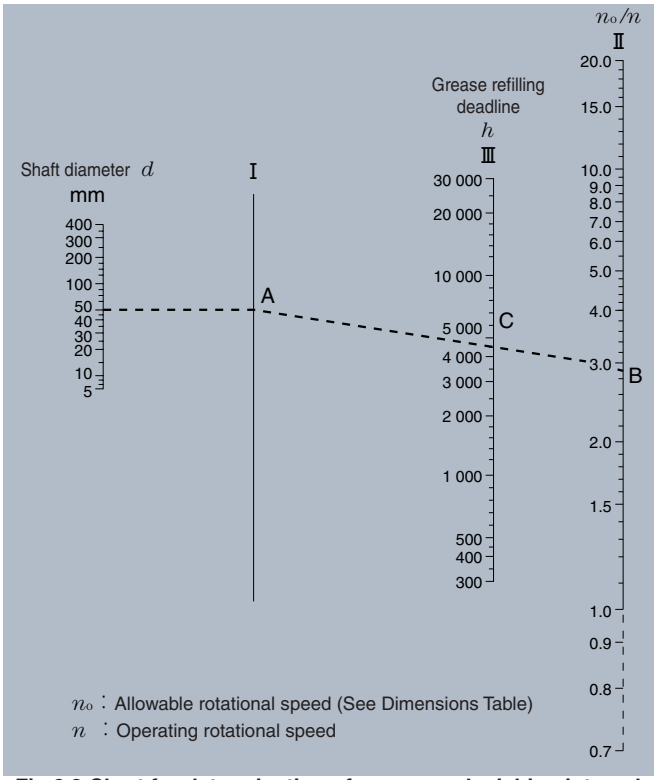


Fig.8.2 Chart for determination of grease replenishing interval

8.2 Oil lubrication

In general, oil lubrication is more suitable for high speed running application or high temperature application than grease lubrication. Oil lubrication is suitable for the case where heat value generating from a bearing or heat value being transferred to a bearing must be discharged outside the bearing.

8.2.1 Lubrication method

(1) Oil bath lubrication

This oil bath lubrication is the most common of the currently available oil lubrication methods, which is applied to bearings running at low or medium speed. An important point in this method is control of oil level in an oil bath.

For that, when bearings are installed on a horizontal shaft, it is common that a point close to the center of the rolling element in the lowest position should be deemed as the oil level to be secured during shutdown. In this case, the housing must be designed in such a profile as to minimize variation in oil level therein. Furthermore, it is desirable to provide the housing with an oil gauge to facilitate level check during running as well as shutdown.

When bearings are installed on a vertical shaft, it is okay if 50 to 80% of the rolling elements are dipped in an oil bath under low speed running, but in the cases of high speed running and bearings used in multiple rows it is desirable to adopt the drip lubrication and circulating lubrication methods, and others described hereunder.

(2) Spray lubrication

This method sprays lubrication oil by an impeller of simple structure, which is mounted on the shaft, without directly dipping a bearing in an oil batch. This can be applied to bearings running at considerably high speed.

(3) Drip lubrication

This lubrication method is used where bearing runs at comparatively high speed and medium and less loads act on thereon. In this method, oil drips from an oiler provided on the top of a bearing unit strikes the rolling elements in its housing for atomizing lubrication (**Fig. 8.3**) or otherwise
a small amount of oil passes through the bearing. In many cases bearing is lubricated with several drips per minute though the number of oil drips per specific unit differs depending on bearing type and dimension.

(4) Circulating lubrication

This circulating lubrication method is adopted to cool down bearings or to lubricate automatically many lubricating portions by a centralized lubrication system. In more detail, as the features of this method the oil feed line is equipped with a cooler to cool down the lubrication oil and, in addition, provided with an oil filter to purify the lubrication oil.

Under this circulating lubrication system, the lubrication oil must securely be discharged from each bearing after having passed through it. For that, it is important to provide an oil inlet and an oil outlet on each bearing in opposite position and to make the oil discharge port size as large as possible or otherwise to discharge the oil compulsorily. (**Fig.8.4**)

(5) Others

Jet lubrication, oil mist lubrication, air-operated oil lubrication, etc. are available as other lubrication methods.

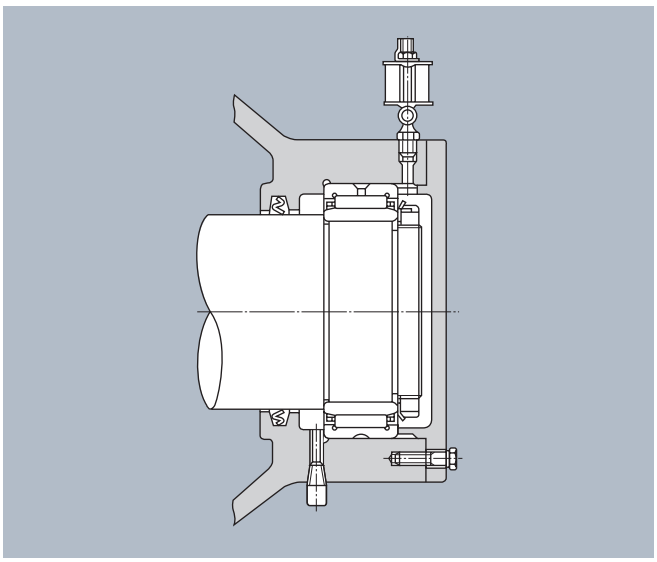


Fig. 8.3 Drip lubrication

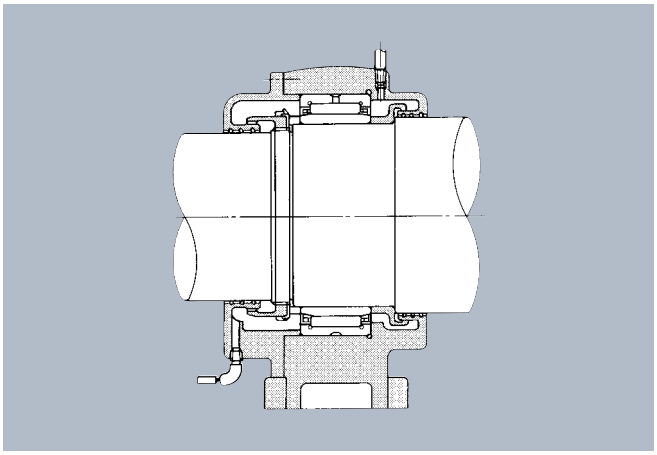


Fig.8.4 Circulating lubrication

8.2.2 Lubrication oil

Mineral oils such as spindle oil, machine oil, turbine oil, etc. are mostly used as the lubrication oil for bearings. However, synthetic oils such as diester oil, silicone oil, fluorocarbon oil, etc. are used under the operating conditions wherein bearings are subjected to running at high temperature of 150˚C and over or low temperature of -30˚C and less.

For lubrication oil, its viscosity is one of the important characteristics that determine the lubrication performance. Too low viscosity of lubrication oil would cause imperfect forming of an oil film and finally cause damage of bearing surface, while too high viscosity of lubrication oil would cause great viscosity resistance, which would then lead to temperature rise and increase of friction loss.

Generally lubrication oil of lower viscosity is used for the faster rotational speed of bearing, while lubrication oil of higher viscosity is used for the heavier bearing load.

Lubrication of bearings needs the oil viscosity every each operating temperature which is specified in **Table 8.4**. **Fig. 8.5** shows the lubrication oil viscosity - temperature characteristic chart, which is referred to in selecting a lubrication oil of optimal viscosity under actual operating temperature.

Furthermore, **Table 8.5** shows the criterion for selection of the lubrication oil viscosity according to the actual bearing operating conditions.

Table 8.4 Oil viscosity required for each bearing type

Bearing type	Required viscosity mm²/s
Radial needle roller bearing	13
Thrust needle roller bearing	20

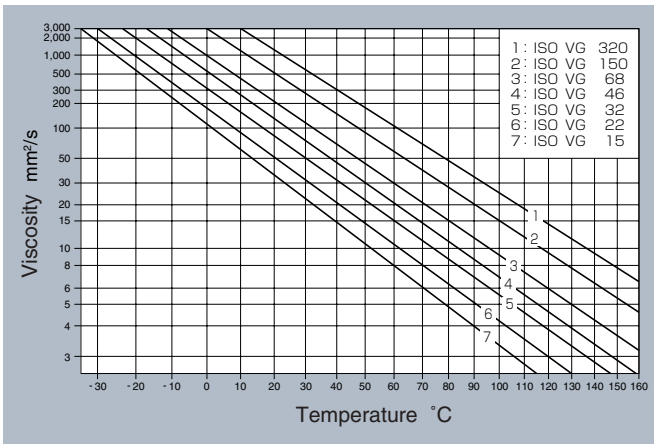


Fig. 8.5 Lubrication oil viscosity - temperature characteristic chart

Table 8.5 Criteria for selection of lubrication oil (Reference)

Bearing operating temperature ˚C	d_n value	ISO viscosity grades for lubrication oil (VG)	
		Ordinary load	Heavy load or shock load
−30～0	up to allowable rotational speed	22 32	46
0～60	up to 15000	46 68	100
	15 000～80 000	32 46	68
	80 000～150 000	22 32	32
60～100	up to 15000	150	220
	15 000～80 000	100	150
	80 000～150 000	68	100 150
100～150	up to allowable rotational speed	320	

Remarks:
1. Subject to oil bath lubrication or circulating lubrication.
2. Apply to **NTN** for other operating conditions other than those specified in this Table.

8.2.3 Oil quantity

Under an forced oil lubrication system, heat value generating from bearing, etc. is equal to the sum of heat value radiated from housing and heat value carried away by lubrication oil.

Where a standard housing is used, the oil lubrication quantity as a guideline can be determined by **formula (8.2)**. The radiated heat value differs depending on the shape /profile of a housing used. It is therefore desirable to determine the lubrication quantity best-suited to an actual equipment by adjusting it from a value approximately 1.5 to 2 times as much as the lubrication rate determined by formula **(8.2)**. In addition, Where the lubrication quantity is calculated assuming that the generated heat value is all carried away by the oil, with no heat radiation from the housing, the oil quantity q can be determined, assuming that the shaft diameter on 2nd vertical line from the right side in (**Fig. 8.6**) Chart for Determination of Oil Quantity q is d=0.

$$Q = K \cdot q \quad \cdots\cdots (8.2)$$

Where,

Q: Oil quantity per bearing cm³/min

K: Coefficient being determined by operating temperature rise of oil (**Table 8.6**)

q : Oil quantity that can be determined from the chart cm³/min (**Fig.8.6**)

8.2.4 Fresh oil refilling interval

Although it depends on actual operating conditions, oil quantity and the kind of lubrication oil used, the fresh oil refilling interval as a guideline is around one year where bearings are lubricated with oil of 50˚C and lower temperature by the oil bath method and is at least 3 months where bearings are lubricated with oil of 80 to 100˚C.

For an important equipment it is desirable to monitor periodically the lubrication performance, cleanliness deterioration, etc. of the lubrication oil in use and to specify the fresh oil refilling interval under the monitor data.

Table 8.6 K-value

Discharged oil temperature - lubricating oil temperature ˚C	K
10	1.5
15	1
20	0.75
25	0.6

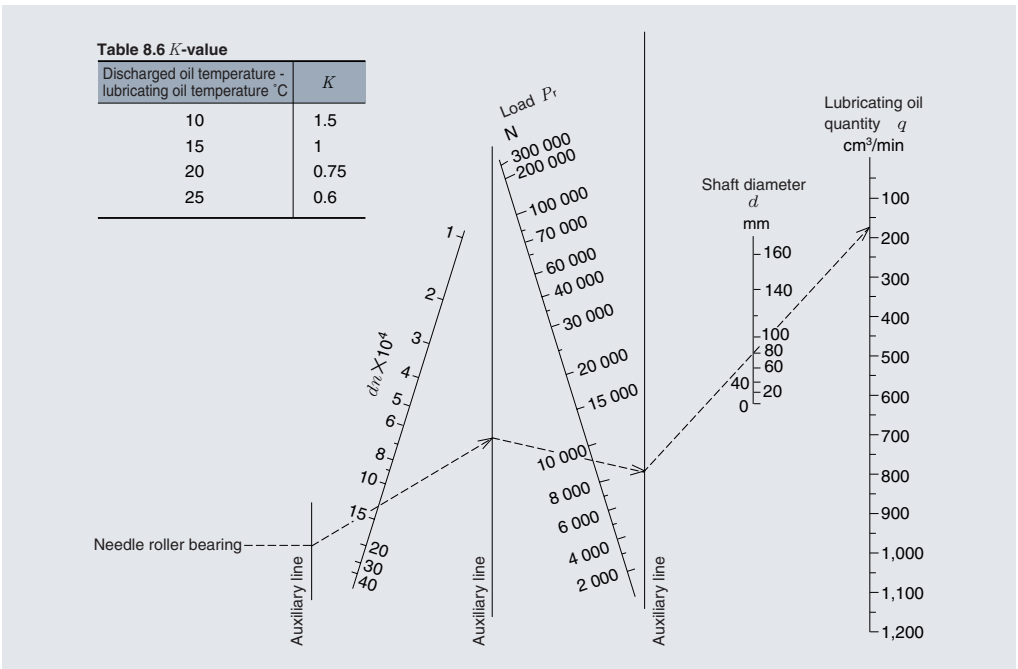


Fig. 8.6 Chart for determination of lubricating oil quantity

9. Sealing Devices

9.1 Non-contact seal and contact seal

The purpose of using a seal is to prevent a lubricant held in a bearing from leaking outside the bearing and to prevent powder, water content, etc. from invading into the bearing from outside.

It is very important to design a sealing device with good care, under full consideration of the operating conditions, lubricating condition, environmental condition, economical merit, etc., so that bearing running is not inversely affected by the sealing device.

The bearing seals are mainly classified into non-contact seal, contact seal, etc. as shown in **Tables 9.1** and **9.2**, which must then be selected correctly according to each application,under full consideration of the characteristics of each sealing type.

Table 9.1 Seals (Non-contact seals)

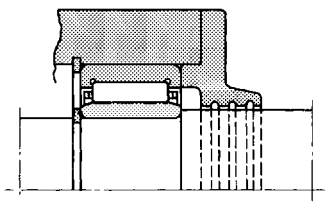
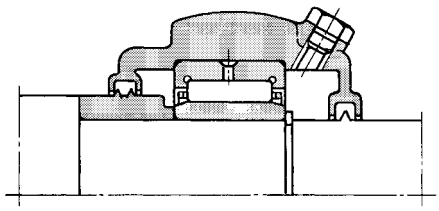
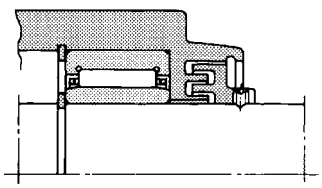
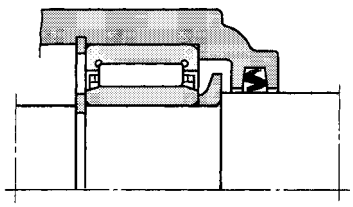
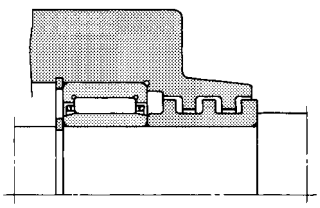
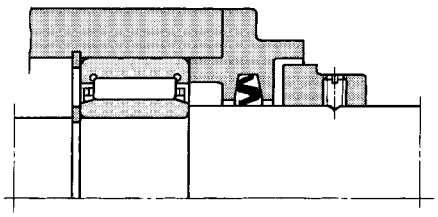
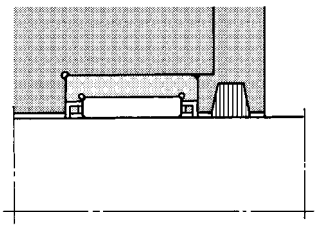
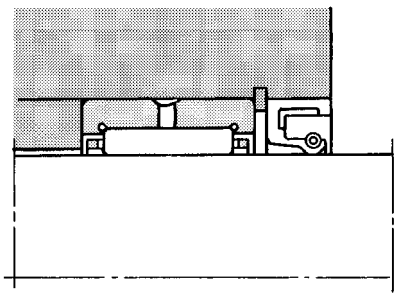
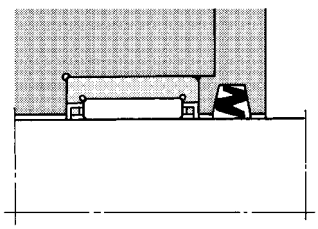
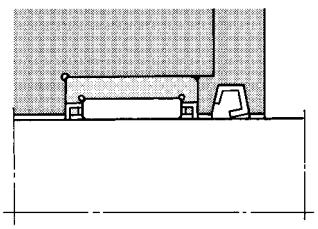
Type	Non-contact seals		
	Seal type using a spacing	Seal type using a centrifuge	Seal type using air pressure
Seal name	Oil groove seal Labyrinth seal (axial, radial)	Slinger seal	Air seal
Features	**<Oil groove seal>** This seal is fitted at either one side of a housing or a shaft, or fitted at the both sides for sealing action. In this case, this seal has also an effect in preventing invasion of foreign matter from outside by holding grease in the oil groove. **< Labyrinth seal >** This seal having a high sealing effect due to its multi-stage labyrinths and long passage is mainly used for grease lubrication. Generally it is suited to high speed bearing running, but it has a dust-proofing effect even under low speed running if the seal grooves are filled up with grease. It is convenient if this seal is provided with a grease nipple.	In oil lubrication, this seal has an effect in slinging and returning the oil flown out along its sleeve by a centrifugal force if its sleeve is provided with projections. A seal example illustrated in **Fig. 9.6** prevents invasion of foreign matter from outside.	———
Application examples	 **Fig. 9.1 Oil groove seal**	 **Fig. 9.4 Slinger with projections**	
	 **Fig. 9.2 Axial labyrinth seal**	 **Fig. 9.5 Slinger intended for back flow of flown-out oil by centrifuge**	
	 **Fig. 9.3 Radial labyrinth seal**	 **Fig. 9.6 Slinger provided at outer side**	

Table 9.2 Seals (Contact seals)

Seal type	Contact seals	
	Seal using mainly direct contact	Seal using mainly fluid lubrication and boundary lubrication
Seal name	Seal ring (felt seal, etc.) O-ring, piston ring	Oil seal, V-shaped ring seal, mechanical seal
Features	This seal type seals a fluid perfectly by pressing its elastic body onto the sliding surface with a constant contact pressure. Generally the contact seals are more excellent in sealing performance than the non-contact seals, but the friction torque and temperature rise thereof are greater than those of the non-contact seals. **<Felt seal>** This is the simplest of the contact seals, which is mainly used for grease lubrication and suited to prevention of fine dust, but oil penetration and outflow are occasionally unavoidable to some extent.	**< Oil seal >** This seal type intended to seal lubricant at the sliding portion between its lip and a shaft has good sealing effect and is an effective seal in the most frequent use. The lip must be oriented outward to prevent invasion of water content and foreign matter from outside and oriented inward to prevent lubricant from leaking out of the housing. Furthermore, another seal type with two or more lips is also available for preventing lubricant leak and dust-proofing individually.
Application examples	 **Fig. 9. 7 Felt seal**	 **Fig. 9.10 Oil seal**
	 **Fig. 9. 8 Z type grease seal**	
	 **Fig. 9. 9 GS type grease seal**	